

Erdős Problem #317

EP-317

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Is there some constant $c > 0$ such that for every $n \geq 1$ there exists some $\delta_k \in \{-1, 0, 1\}$ for $1 \leq k \leq n$ with $0 < \left| \sum_{1 \leq k \leq n} \frac{\delta_k}{k} \right| < \frac{c}{2^n}$? Is it true that for sufficiently large n , for any $\delta_k \in \{-1, 0, 1\}$, $\left| \sum_{1 \leq k \leq n} \frac{\delta_k}{k} \right| > \frac{1}{[1, \dots, n]}$ whenever the left-hand side is not zero?

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 Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
 Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>